

HabitAt: Bridging Humans and Wildlife toward a Sustainable Future

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Abstract

Urban pollution poses significant challenges to both human and wildlife health, necessitating innovative approaches to promote sustainable coexistence. This study explores the potential of human-animal interaction as a lens to foster environmental awareness and empathy. We propose HabitAt, a conceptual design prototype that leverages animals' superior sensory capabilities to detect urban pollution, enabling a deeper understanding of environmental conditions while encouraging sustainable behavior. To inform the design, we conducted a participatory workshop where participants engaged in role-playing activities to experience urban environments from multiple angles. Observations of the workshop were synthesized into HabitAt's design, which integrates ecological data visualization and interactive elements to create deeper human-animal connections and provoke daily actions in environmental protection. Moving forward, we will refine app features, resolve technical challenges, and expand its applications to better support sustainable urban development.

CCS Concepts

• **Human-centered computing** → **Interaction design**.

Keywords

Sustainable Cities and Communities, Human-Wildlife Coexistence, Participatory Workshop, Empathy-Driven Interaction Design

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1 Introduction

1.1 Urban Environmental Challenges for a Sustainable Future

Urbanization and industrialization have intensified environmental pollution, posing severe risks to health and ecosystems. Pollutants

such as air and noise pollution contribute to digestive disorders and organ damage [21]. In line with the 11th Sustainable Development Goal (SDG) of "making cities and human settlements inclusive, safe, resilient, and sustainable" [15], there is an urgent need to address these environmental issues to create a greener urban future. Protecting human health and ecosystems requires not only large-scale policies but also conscious actions in daily life. Research suggests that fostering environmental awareness is key to promoting sustainable behaviors, echoing UNESCO's call to integrate sustainability into education [11]. To promote sustainable daily actions, raising awareness of health and environmental quality seems to be essential for fostering long-term behavioral change.

1.2 Wildlife in the City—the Hidden Victims of Urban Pollution

Urban areas are home not only to humans but also to a variety of animal species. In cities like Shanghai, where this study is conducted, wildlife such as bats, hedgehogs, and raccoon dogs coexist with the human population. However, urban wildlife may be much more vulnerable to pollution than humans. Research has shown that urban pollutants, such as noise and artificial light, can affect wildlife through multiple sensory channels, ultimately altering their perceptions and behaviors [6]. While these pollutants typically pose long-term health risks to humans, they may have more immediate and severe impacts on urban wildlife, sometimes even leading to their death. For example, research has indicated that artificial lighting can disorient turtles, causing them to move toward nearshore buildings and roads instead of the ocean, which puts their lives in danger [10]. Unfortunately, there is often a lack of public awareness concerning urban wildlife and how daily pollution impacts their health and habitats.

1.3 From Threatened Habitat to Sustainable HabitAt

As we confront the intricate urban environmental challenges, it is crucial to recognize that we share our cities and ecosystems with wildlife. Though both humans and animals face the challenges of urban pollution, their sensory experiences and perceptions differ, offering unique insights into the urban environment. By leveraging these differences, we find a great opportunity to develop innovative approaches to environmental protection. In response, we present HabitAt, an app designed to bridge the sensory gap between humans and wildlife in urban spaces. By utilizing animals' heightened sensitivity, HabitAt helps users detect pollution sources and understand wildlife behavior in real time. This approach fosters empathy,

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strengthens human-wildlife relationships, and encourages sustainable behaviors. This paper explores the theoretical foundations, workshop insights, and the conceptual development of HabitAt, highlighting its potential for a more sustainable urban future.

2 Related Theories and Work

Considering the role of animals' sensory systems, we explore the ecological concept of bioindicators—living organisms used to screen ecosystem conditions [17]. Bioindicators are widely employed to monitor environmental health, particularly for pollution detection and ecosystem evaluation. For example, the diversity of aquatic animals, such as benthic macroinvertebrates, serves as a reliable bioindicator of water quality, with changes in species diversity reflecting the improvement or degradation of water conditions [19]. Despite their proven role in pollution assessment, bioindicators are largely confined to specialised applications. This limitation raises an opportunity to incorporate the bioindicator concept into our design, inspiring to make urban environmental detection more intuitive and accessible for daily use.

On exploring the relationship between humans and animals, the Biophilia Hypothesis provides a valuable theoretical lens. This hypothesis posits that humanity has an innate affinity for life forms and the natural world, which has the potential to foster psychological well-being and heighten environmental awareness and responsibility [12]. In the design field, many interactive products align with this concept by simulating connections with living creatures. For example, the classic Tamagotchi engages users by simulating the needs and growth of a virtual pet, fostering emotional attachment [16]. Similarly, Sony Aibo, an AI-powered robotic pet, constructs its interactive connection with users by providing enhanced emotional support, extending its applications beyond home settings to therapeutic uses in hospitals [25]. However, these designs primarily focus on the owner-pet relationship instead of the dynamic between humans and animals. Noticing this gap, we draw inspiration from the game “Tabikaeru”, in which a virtual frog travels and sends back photos, illustrating a balanced and reciprocal human-animal relationship [2].

Together, these perspectives illuminate opportunities to integrate ecological insights and redefine human-animal relationships through a more inclusive and participatory way. These insights shape our workshop-based user research and the conceptual framework of our project.

3 Workshop-Based Research

3.1 Purpose of Our Workshop

Motivated by the pressing challenges in environmental awareness and sustainability, we pose the following research questions:

- (1) How can we enhance public awareness and sensitivity toward environmental issues?
- (2) How can we foster empathy between humans and urban wildlife to promote sustainable coexistence?

To explore these questions, we conduct a participatory workshop as a case study to generate insightful findings for future research and design practices.

3.2 Theoretical Support

We draw on key theoretical frameworks to inform our workshop design.

“We must begin in empathy, by becoming the animals before we can save them.” This idea, proposed by David Sobel, highlights the role of empathy in environmental education [22]. Research by P. Wesley Schultz further reflects that participants who were instructed to adopt the perspective of an animal suffering from pollution score better in environmental concerns than those who did not [18]. This highlights the power of perspective-taking in fostering empathy and responsibility for the natural world. Building on this foundation, we incorporated the Role-Taking Theory, which emphasizes adopting the perspectives of others as a method to understand their experiences and challenges [13]. Jenny L. Davis extends this framework within participatory and co-design methodologies in HCI, highlighting its relevance in addressing marginalized groups and cross-species interactions [4].

Based on these theoretical insights, our workshop includes two role-taking exercises. First, participants will take the role of “urban wildlife” to deeply experience their perspectives of urban pollution. Then, participants will act as animal journalists, using AI to craft news stories that amplify their voices and highlight urgent environmental crises.

Moreover, inspired by the concept of place-based education, which emphasizes learning within familiar local contexts to foster meaningful connections [23], we selected four common wild animals in Shanghai—squirrels, bats, hedgehogs, and raccoon dogs—as our focus species. This choice was driven by two key factors, based on *What to Do with Common Urban Wildlife in Shanghai-Community Management Manual* [20] and other literature [1, 3, 5, 7, 8, 14, 24]. First, while these animals actively adapt to urbanization, they still suffer from its impact. Second, their highly sensitive senses enable them to serve as biological detectors.

3.3 Workshop Procedure

The workshop, titled *Where Are the Animal Journalists?*, consisted of five steps that simulated the process of becoming a “qualified” animal journalist through role-taking strategies (Figure 1). It guided participants through a process of first observing their environment as humans, then adopting animal perspectives to reassess their observations, and finally returning to human perspectives to advocate for animals. To ensure diversity and depth, we selected four graduate participants from Tongji University with different academic backgrounds, varying degrees of environmental awareness, and distinct experiences with animals. The five steps of the workshop are illustrated as follows:

3.3.1 Immersion in Environment: Journey of Pollutant Discovery from a Human Perspective. Participants took photos to document the environmental conditions on their way to the workshop, fostering conscious observation from a human perspective.

3.3.2 Knowledge Learning: Introduction to Urban Pollution and Wildlife. Participants were introduced to detailed profiles of four common urban wildlife species in Shanghai—squirrels, bats, hedgehogs, and raccoon dogs—to deepen their awareness of these animals' living conditions in polluted urban environments.

Workshop Procedure	Role-taking
1 Immersion in Environment: Journey of Pollutant Discovery from a Human Perspective	HUMAN
2 Knowledge Learning: Introduction to Urban Pollution and Wildlife	WILDLIFE
3 Zine Creation & Empathy: Retracing the Journey from an Animal's Perspective	HUMAN
4 Discussion and Advocacy: Becoming an Animal Journalist	HUMAN (As Animal Journalist)
5 Post-workshop interviews: Reviewing, Reflecting and Suggesting	HUMAN

Figure 1: Diagram illustrating the five steps of the workshop and the roles played by the participants in each step.

3.3.3 *Zine Creation with Empathy: Retracing the Journey from an Animal's Perspective.* Participants selected one species to “transform” into and created zines retracing their journeys to the workshop from the animal’s perspective. Drawing from Kafka’s Metamorphosis, they began with the iconic opening line: “One morning, as I was waking up from dreams, I discovered that in bed I had been changed into a ...” [9], using collages, sketches, and words to express their emotions. Figure 2 displays the zines they created.

3.3.4 *Discussion and Advocacy: Becoming an Animal Journalist.* Participants acted as animal journalists, using AI tools to craft news articles about the environmental challenges faced by their chosen species. In an editorial meeting, they discussed the significance of their stories and collaboratively designed a newspaper titled *Invisible Citizens of the City* (Figure 2). Participants received animal journalist certifications in recognition of their advocacy efforts.

3.3.5 *Post-Workshop Interviews: Reviewing, Reflecting, and Suggesting.* Participants shared their reflections in post-workshop interviews in terms of environmental conservation and the sustainable coexistence of humans and wildlife.

3.4 Observations and Findings of the Workshop

After the workshop, multimodal materials including videos, pictures, and design artifacts (zines and newspapers) were collected and analyzed.

We observed that the workshop significantly increased participants’ awareness of urban wildlife. One noted, “I seldom noticed urban wildlife before. People see the cute side, but their struggles remain invisible.” Another shared, “Noise pollution has always bothered me, but seeing it from a hedgehog’s perspective made me want to take action.” These reflections highlight the power of empathy-driven design. Additionally, analysis of the zines and newspapers created during the workshop showed participants’ critical engagement with urban wildlife and pollution issues.

Post-workshop interviews provided deeper insights into human-wildlife relationships. All participants agreed that understanding



Figure 2: Pictures of zines and newspapers made by participants in the workshop.

urban wildlife reduces fear of coexistence and fosters empathy. One participant noted, “People see urban wildlife as pets, but animals like raccoon dogs are the true natives of Shanghai.” This perspective inspired us to embrace post-humanist thinking, recognizing urban wildlife as equal citizens rather than subordinates.

4 Design of HabitAt

4.1 Overview

Combining “Habit” and “At”, HabitAt aims to integrate daily environmental actions into place-based contexts, fostering curiosity, empathy, and ecological responsibility. HabitAt is a mobile and smartwatch application that integrates gamified elements to enhance user engagement. It provides real-time feedback on urban pollution using device sensors like GPS and light sensors, allowing users to mark local pollution sources while interacting with virtual wildlife companions that symbolize local species.

HabitAt also extends beyond the app, serving as a system that connects users, local communities, and urban environmental organizations (Figure 3). This system collects user feedback on urban pollution, channels insights to relevant organizations, and engages users with localized wildlife education. Through fostering universal participation, HabitAt empowers individuals to contribute to urban pollution management and supports the collective exploration of a sustainable urban future.

4.2 Key Features

4.2.1 *Mobile Application.* The HabitAt mobile app offers a digital comfort space as the home page where users interact with a virtual wildlife companion (Figure 4). Based on local urban species, the AI-driven companions provide visual feedback, simulating wildlife in urban environments. It engages users in conversations about real-time pollution data, weather, and other relevant factors. Users can invite different animals for diverse interactions; however, the animals may refuse to participate if the pollution levels in the users’ contexts exceed acceptable thresholds.

The core features of the app are accessed through interactive elements within the digital space. By clicking graphics of the window,

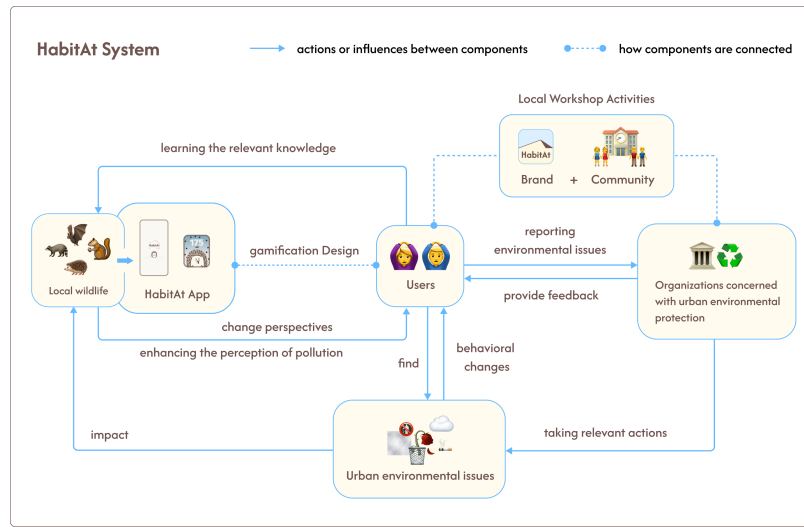


Figure 3: The HabitAt system, including HabitAt’s relationship with local wildlife, users, urban environments, and related organizations.

door, map on the wall, and letters on the table, users can navigate to different pages with relevant functions (Figure 4).

My Window: By clicking the window on the home page, users will be navigated to *My Window* where they can get real-time environmental information through data on weather, air quality, light pollution, and noise exposure levels.

My Marks: By clicking the map on the wall, users can go to *My Marks* where they can view, mark, and edit pollution incidents.

My Neighbors: Swiping right on the home page and clicking the door, users can “step outside” to *My Neighbors*, where they can interact with animal neighbors, invite them to home and view their profiles.

My Letters: Swiping right and clicking the letters on the table, users will be taken to *My Letters* where they can receive AI-generated newspapers and exchange letters with animals and relevant organizations to further report and track pollution.

4.2.2 Smartwatch Application. The HabitAt smartwatch application is specifically designed for users' daily commutes, seamlessly integrating with the mobile app to serve as a comprehensive ecosystem. Its main features include:

Virtual Animal Commute Companions and Personalized Suggestions: During users' commutes, the animal in the mobile application will be displayed on the watch, accompanying users through their daily activities and providing personalized health recommendations and behavioral guidance.

Real-time Perception and Quick Marking of Pollution: The smart-watch app utilizes multiple sensors to capture and display real-time pollution data. It detects pollution from the lens of the chosen animal companion during commutes and provides users a quick process to mark, track and address urban environmental issues. Figure 5 presents a series of HabitAt's watch interfaces when light pollution is detected and marked.

4.3 Incorporating AI Agents into the HabitAt System

In the HabitAt system, two types of AI agents have been designed to provide intelligent assistance (Figure 6).

The first type is the virtual wildlife companion, which provides personalized health and behavioral recommendations across devices, such as the generation of context-aware interfaces on smart-watch and the cross-species letter exchanges on smartphone.

The second type is the intelligent news agency which utilizes AI to generate news content and layouts based on local pollution data and users' commuting conditions.

Besides working individually, the two agents exchange information to provide users with a more precise, personalized, and contextual feedback.

5 User Evaluation

We evaluated HabitAt’s usability and functionality with 12 participants (aged 20–30, all residing in Shanghai, some from the previous workshop). The user testing process involved three tasks: (1) exploring the HabitAt prototype, (2) completing a questionnaire incorporating the UEQ and conceptual questions, and (3) participating in semi-structured interviews for deeper insights.

Overall, HabitAt received positive feedback on its usability. Attractiveness scored 5.75, indicating moderate visual and functional appeal, while perspicuity scored 6.21, demonstrating ease of use. However, efficiency scored the lowest at 5.29, highlighting areas for optimization, such as navigation improvements. Interviews revealed that first-time users struggled with interaction guidance. Some were unsure where to click, fearing they might miss important content. To address this, a structured onboarding process will be implemented in future iterations.

In terms of functionality, participants found HabitAt engaging and informative, especially appreciating the portrayal of animals as urban citizens. They also agreed with the sustainability values

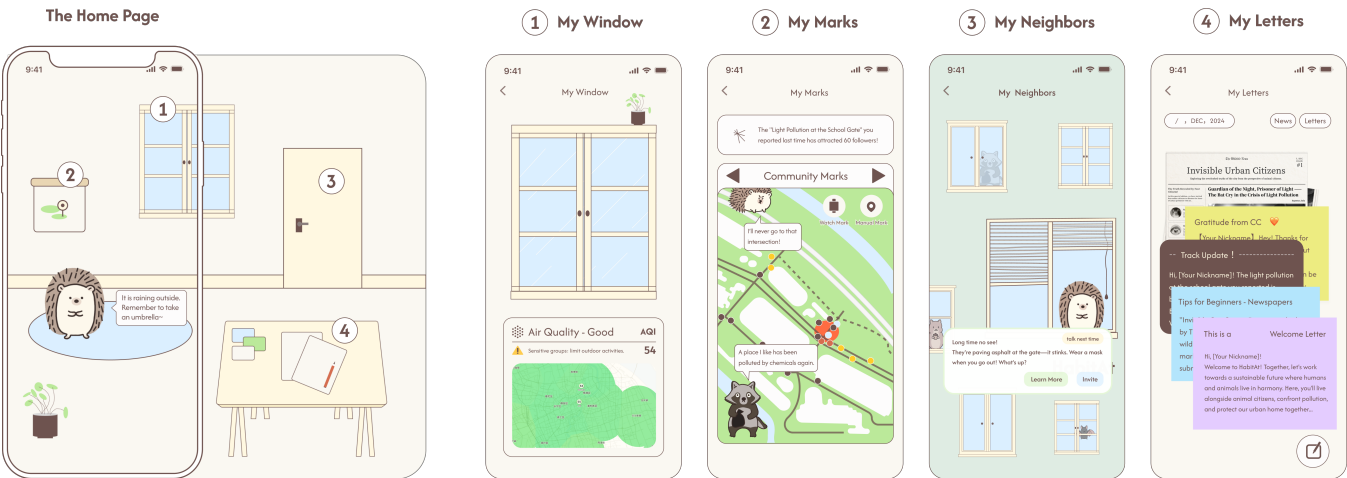


Figure 4: High-fidelity prototypes of the HabitAt mobile version. Wireframes of five interfaces: Home, My Window, My Marks, My Neighbors, and My Letters. Connections between elements on the home page and their corresponding interfaces are indicated using numbered icons.

A Case of Marking Light Pollution in HabitAt on Watch

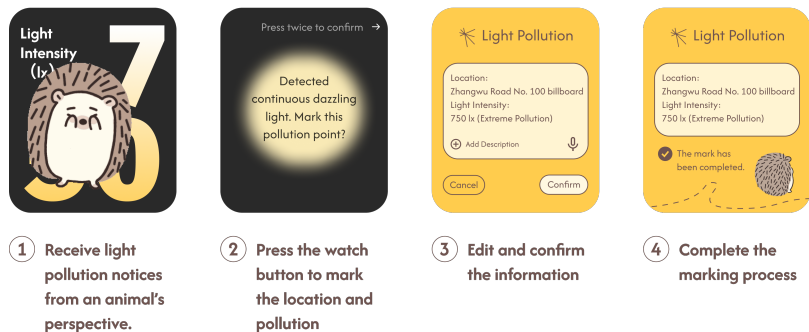


Figure 5: High-fidelity prototypes of the HabitAt smartwatch version, showing four interfaces that illustrate the steps to mark urban pollution.

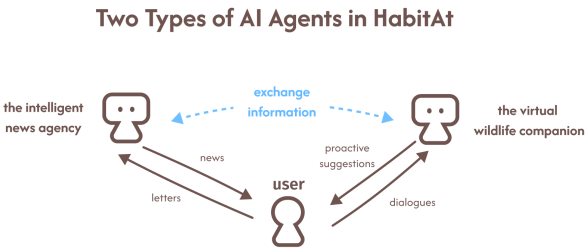


Figure 6: The relationship between users and two types of AI agents in HabitAt.

delivered by HabitAt, expressed a willingness to take more actions toward achieving a sustainable urban future. However, a common

suggestion was to deepen the connection between virtual wildlife companions and wildlife in reality, ensuring that users' environmental actions within HabitAt make an actual contribution to real urban habitats. Future iterations will explore more ways to bridge digital engagement with diverse and deeper offline actions.

6 Conclusion & Future Work

This study highlights the pressing environmental challenges in urban areas and the often-overlooked impact of pollution on urban wildlife. Based on rich literature exploration, we employs a participatory workshops including role-playing sessions, fostering empathy and perspective-taking. Through our conceptual design of HabitAt, we propose an interactive platform that bridges human and wildlife perceptions and experiences, encouraging awareness of environmental responsibility and actions in daily life toward a sustainable urban future.

Future work will refine efficiency through adding an onboarding feature and explore connecting digital and real-world interactions. Additionally, challenges such as user privacy concerns in multi-sensor data collection, AI-driven content generation complexity, and difficulties in measuring behavioral impact will be addressed.

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